

Ziya Sherzeen S

Artificial Intelligence and Data Science Engineer

Education

Year	Degree	Institute Name	Marks
2022-2026	B. Tech (Artificial Intelligence and Data Science)	Karpagam Institute of Technology	8.34

Contact Info

Mail: zsherzeen@gmail.com

Leetcode: https://leetcode.com/u/Ziya_sherzeen/

LinkedIn: <https://www.linkedin.com/in/ziya-sherzeen/>

GitHub: <https://github.com/ZiyaSherzeen>

Career Objective

An Artificial Intelligence and Data Science engineering student with a strong foundation in Machine Learning and full-stack development. Passionate about crafting intelligent, data-driven solutions that integrate analytical depth with practical web technologies. Eager to contribute to innovative projects that solve real-world problems through the synergy of AI and modern software engineering.

Projects

1.Vehicle Routing Problem | Guided Project (Jan 8-Jan 9)

Objective: Optimized the Vehicle Routing Problem using Genetic Algorithms in DEAP, minimizing travel distance and time.

- Developed a solution for the Vehicle Routing Problem to optimize route planning for multiple vehicles.
- Utilized Genetic Algorithms in DEAP with a distance matrix and time-based fitness evaluation.
- Achieved minimized travel time and distance, ensuring efficient and effective routing.

2.Dimensionality-Reduction-Using-K-Means-for-Activity-Recognition | Guided Project (Jan 10)

Objective: Implemented K-Means clustering for dimensionality reduction in activity recognition, improving feature selection and model efficiency.

- Used K-Means clustering to identify and group redundant features in the activity recognition dataset.
- Selected key features by clustering and identifying the most informative ones.
- Trained a model using the reduced feature set, improving computational efficiency and maintaining accuracy.

3.Market Segmentation Using DBSCAN | Guided Project (Mar 9)

Objective: Segment customers based on spending behaviour and purchase frequency using DBSCAN, helping businesses with targeted marketing and customer retention.

- DBSCAN dynamically determines clusters instead of requiring a fixed number.
- Effectively segments **diverse customer groups** without assuming uniform distribution.
- Helps businesses **personalize offers, optimize marketing, and improve customer engagement** based on spending habits.

4. NutriTwin - AI-Powered Indian Nutrition Intelligence Platform | Ongoing Project (April,2025-Present)

Objective: Developed an AI-powered nutrition platform for Indian diets using food recognition and LLMs.

- Built **CLIP + EfficientNet-B4** food recognition with LLM-based dish and portion analysis.
- Integrated **INDB, IFCT, USDA & Open Food Facts (~500K entries)** for nutrition data retrieval.
- Implemented **deficiency tracking, RLHF correction, and context-aware modes** using Claude/Groq APIs.

Certificates

- Certified AWS Academy Cloud Foundations Graduate
- Certified in RDBMS PostgreSQL through the Spoken Tutorial Project by IIT Bombay, upon successful completion of the training and online examination
- Completed the MongoDB certification for Document Modelling on the official MongoDB platform.

Skills

Programming Languages: Python, Java, C

Web Development: HTML, CSS, JavaScript, React Js

Machine Learning: KNN, Genetic Algorithm, DEAP, NLP, DBSCAN

Soft Skills: Critical Thinking, Team Collaboration, Communication Skills, Leadership

Achievement

Selected for Ideathon Round – Girl Hackathon 2025 by Google

Qualified for Round 2 of Google's nationwide Girl Hackathon (SWE Track), based on performance in the initial selection round (Problem-Solving)

Selected for Niral Thiruvizha 3.0 – TNSDC | Naan Mudhalvan

Selected for the statewide innovation hackathon and successfully completed the **Regional Review** stage.